
3AL Tutorial 12 2024 Solutions

1. For each of the given rings R and subsets S , prove that S is a subring of R .

- (a) Let $R = M_2(\mathbb{R})$ and $S = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R}, a + c = b + d \right\}$.
- (b) Let R be an arbitrary ring and $a \in R$, define $S = \{r \in R : ar = 0\} \cup \{1\}$.
- (c) Let $R = \mathbb{Z} \times \mathbb{Z}$ and $S = \{(x, y) \in R : x - y = 0\}$.
- (d) Let $R = \mathbb{R}$ and $S = \{x + \sqrt{3}y : x, y \in \mathbb{Z}\}$.

(a) We have $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in S$ since $1 + 0 = 1 = 0 + 1$. Given $\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \in S$, we have

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} - \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} a - a' & b - b' \\ c - c' & d - d' \end{pmatrix} \in S$$

since

$$a - a' + c - c' = (a + c) - (a' + c') = (b + d) - (b' + d') = b - b' + d - d'.$$

Furthermore,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} aa' + bc' & ab' + bd' \\ ca' + dc' & cb' + dd' \end{pmatrix} \in S$$

since

$$\begin{aligned} aa' + bc' + ca' + dc' &= (a + c)a' + (b + d)c' = (a + c)(a' + c') \\ ab' + bd' + cb' + dd' &= (a + c)b' + (b + d)d' = (a + c)(b' + d') = (a + c)(a' + c'). \end{aligned}$$

Thus by the subring test, S is a subring of R .

- (b) It is clear that $1 \in S$. Now, let $r, s \in S$. Then $a(r - s) = ar - as = 0 - 0 = 0$ and so $r - s \in S$. Furthermore, $a(rs) = (ar)s = 0$ and so $rs \in S$. Thus S is a subring of R .
- (c) Note that $(1, 1)$, the identity of $\mathbb{Z} \times \mathbb{Z}$ is in S since $1 - 1 = 0$. Further, notice that all elements of S have the form (x, x) where $x \in \mathbb{Z}$ (this step is not necessary,

but it simplifies our notation a bit). Let $(x, x), (y, y) \in S$. Then $(x, x) - (y, y) = (x - y, x - y)$, so $(x, x) - (y, y) \in S$. Finally, $(x, x)(y, y) = (xy, xy) \in S$. So S is indeed a subring of $\mathbb{Z} \times \mathbb{Z}$.

- (d) We have $1 = 1 + \sqrt{3} \cdot 0 \in S$. Let $x + \sqrt{3}y, x' + \sqrt{3}y' \in S$ where $x, x', y, y' \in \mathbb{Z}$. Then

$$(x + \sqrt{3}y) + (x' + \sqrt{3}y') = (x + x') + \sqrt{3}(y + y') \in S,$$

since $x + x', y + y' \in \mathbb{Z}$. Furthermore,

$$(x + \sqrt{3}y)(x' + \sqrt{3}y') = (xx' + 3yy') + \sqrt{3}(xy' + x'y) \in S,$$

since $xx' + 3yy', xy' + x'y \in \mathbb{Z}$.

2. Let R be a ring with subrings S_1 and S_2 . Prove or disprove the following:

- (a) $S_1 \cup S_2$ is a subring of R .
- (b) $S_1 \cap S_2$ is a subring of R .
- (c) $S_1 \times S_2$ is a subring of $R \times R$.

- (a) False. Consider $R = \mathbb{R}$ and $S_1 = \mathbb{Q}$ and $S_2 = \{x + \sqrt{3}y : x, y \in \mathbb{Z}\}$. We have $\frac{1}{2} \in S_1$ and $\sqrt{3} \in S_2$. But $\frac{1}{2} + \sqrt{3} \notin S_1 \cup S_2$.
- (b) True. Since $1 \in S_1, S_2$, we have $1 \in S_1 \cap S_2$. For $a, b \in S_1 \cap S_2$, we have $a - b, ab \in S_1, S_2$ since S_1 and S_2 are both subrings. Thus $a - b, ab \in S_1 \cap S_2$.
- (c) True.

3. Let R and S be rings.

- (a) Prove that $R \times S$ is a ring.
- (b) Prove that $R \times S$ is commutative if and only if R and S are both commutative.

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- (a) Routine calculations. Note that the zero is $(0_R, 0_S)$ and the unity is $(1_R, 1_S)$.
- (b) Let $(r, s), (r', s') \in R \times S$. We have $(r, s)(r', s') = (r', s')(r, s)$ if and only if $(rr', ss') = (r'r, s's)$, that is, $rr' = r'r$ and $ss' = s's$, i.e. R and S are commutative.

4. Let R be a ring and $r \in R$. Prove that if r is a unit, then r cannot be a zero divisor.

Suppose to the contrary, that r is a zero divisor. Then there exists $s \in R$ with $s \neq 0$ such that $rs = 0$. Then we have $r^{-1}rs = r^{-1}0$ and so $s = 0$, a contradiction. Thus r cannot be a zero divisor.

QUESTION: If r is a zero divisor, can it be a unit?

5. Determine whether the following statements are true or false:

- (a) The trivial ring is commutative.
- (b) The trivial ring is a subring of every ring.
- (c) The subring of a domain is a domain.
- (d) The subring of a division ring is a division ring.

NOTE: The trivial ring is $\{0\}$ where $0 + 0 = 0$ and $0 \cdot 0 = 0$. The element 0 is both the zero and the identity.

- (a) The trivial ring is commutative, this is obvious since we only have one possible multiplication $0 \cdot 0 = 0$.
- (b) The trivial ring is in general not a subring of every ring since it will not contain that larger ring's identity.
- (c) The subring of a domain is a domain. Let R be a ring and S a subring. If $a, b \in S$ with $ab = 0$, then $a = 0$ or $b = 0$ since $S \subseteq R$ and R is a domain. Thus $a = 0$ or $b = 0$ and so S is also a domain.

(d) The subring of a division ring is not necessarily a division ring. Consider $\mathbb{Z} \subseteq \mathbb{R}$.

6. Let R and S be rings.

(a) Suppose $R \times S$ is an integral domain.

(i) Prove that at least one of R and S is an integral domain.

HINT: Proof by contradiction.

(ii) Assume, without loss of generality, that R is an integral domain. Prove that $S = \{0\}$.

(b) Prove that if R is an integral domain and $S = \{0\}$ then $R \times S$ is an integral domain.

(c) Find a counterexample to the following statement: If R and S are fields, then $R \times S$ is a field.

(a) (i) Suppose neither R and S are integral domains. Then there exist nonzero $r, r' \in R$ and $s, s' \in S$ such that $rr' = 0$ and $ss' = 0$. Then $(r, s)(r', s') = (rr', ss') = (0, 0)$, where neither (r, s) nor (r', s') are zero. This contradicts $R \times S$ being an integral domain. So, we must have that at least one of R and S is an integral domain.

(ii) Let $s \in S$. We need to show, using our above results that s must be 0. We have $(1, 0), (0, s) \in R \times S$. Then $(1, 0)(0, s) = (0, 0)$. But $R \times S$ is assumed to be an integral domain, so this is only possible if $(1, 0) = (0, 0)$ or $(0, s) = (0, 0)$. We cannot have $1 = 0$ in R since it is an integral domain. So, we must have $s = 0$ for all $s \in S$. Thus $S = \{0\}$.

(b) Let $(r, s), (r', s') \in R \times S$. Suppose $(r, s)(r', s') = (0, 0)$. Then $rr' = 0$ and $ss' = 0$. We have $s' = s = 0$ since $S = \{0\}$. Since R is an integral domain, we have $r = 0$ or $r' = 0$. Thus $(r, s) = (0, 0)$ or $(r', s') = (0, 0)$ and so $R \times S$ is an integral domain.

(c) Consider $R = S = \mathbb{R}$. The element $(1, 0)$ does not have a multiplicative inverse in $\mathbb{R} \times \mathbb{R}$. Thus $\mathbb{R} \times \mathbb{R}$ is not a field, while $\mathbb{R}$ is a field.

7. A ring R is called **Boolean** if $x^2 = x$ for all $x \in R$.

- (a) Prove that, in a Boolean ring, $r + r = 0$ for all $r \in R$.

HINT: Take $x = r + r$ and apply the identity $x^2 = x$.

- (b) Show that, in a Boolean ring, $r = -r$ for all $r \in R$.

- (c) Prove that every Boolean ring is commutative.

HINT: Take $x = r + s$ and apply the identity $x^2 = x$, use the previous results too.

- (a) Let $r \in R$. By applying the identity to $x = r + r$, we obtain $(r + r)^2 = r + r$. Expanding the left-hand side, we get $r^2 + 2r + r^2 = r + r$. But $r^2 = r$, so $2r = 0$ and thus $r + r = 0$.

- (b) This follows immediately from $r + r = 0$.

- (c) Let $r, s \in R$. We need to show $rs = sr$. Consider $r + s \in R$. We have

$$(r + s)^2 = (r + s)(r + s) = r^2 + sr + rs + s^2 = r + sr + rs + s$$

But $(r + s)^2 = r + s$. So we have $r + s = r + sr + rs + s$ and so $0 = sr + rs$, thus $sr = -rs = rs$. Thus R is commutative.

8. Let R be a ring. Suppose that $x^2 - x \in Z(R)$ for all $x \in R$.

- (a) Prove that $sr + rs \in Z(R)$ for all $r, s \in R$.

HINT: Take $x = r + s$ and use the fact that $x^2 - x \in Z(R)$.

- (b) Prove that $r^2 \in Z(R)$ for all $r \in R$ and conclude that R is commutative.

HINT: $Z(R)$ is a subring of R , so it is closed under addition.

- (a) Let $r, s \in R$. Consider $r + s \in R$, we have

$$(r + s)^2 - (r + s) = r^2 + sr + rs + s^2 - r - s \in Z(R).$$

Since $r^2 - r, s^2 - s \in Z(R)$, we have $sr + rs \in Z(R)$.

(b) Let $r, s \in R$. Since $sr + rs \in Z(R)$, we have $r(sr + rs) = (sr + rs)r$. Thus, $rsr + r^2s = sr^2 + rsr$ and hence $r^2s = sr^2$. Thus $r^2 \in Z(R)$. Finally, $r^2 - (r^2 - r) = r \in Z(R)$. Since this holds for all $r \in R$, we must have $Z(R) = R$ and so R is commutative.